
Historical Information for Home Credit Ranking: A Fixed-Pipeline Development Comparison

JIANG Zerui, CHEN Kepeng, MA Ruiyang, WU Ziyi

Master's students, DDM programme

The Hong Kong University of Science and Technology

9 October 2026

Abstract

Credit-risk models combine application information with records of previous borrowing and repayment, but their separate contributions are often obscured by model tuning. We examined the incremental value of joined credit histories in the Home Credit dataset using logistic regression, random forest and LightGBM. Each pipeline retained its original settings across three information sets evaluated on common development folds. Basic historical aggregation and 39 additional behavioural features increased ROC AUC and average precision in every paired fold. Total mean-fold AUC gains were 0.021006, 0.014248 and 0.023029, respectively. A separate bounded search increased LightGBM mean AUC from 0.780230 to 0.782349 through configuration refinement and rank fusion. Additional candidate feature groups failed the adoption rule. These retrospective development results favour evaluating information additions within a specified pipeline, while showing smaller subsequent tuning gains in this study. Selection dependence, a previously viewed holdout and different cross-validation and deployment fusion rules limit claims about generalisation.

1 Introduction

Payment-difficulty ranking uses information available at application time. Application attributes describe the applicant and current request; previous loans, balances and repayments may supply complementary evidence. Home Credit provides both sources [1]. A final score alone cannot distinguish gains from historical information from gains due to the modelling procedure.

We compared logistic regression (LR) with Elastic-Net regularisation [2], random forest (RF) [3] and LightGBM (LGB) [4]. They offer signed coefficients, randomised-tree ensembles and sequential boosting, respectively. Our LR includes nonlinear basis terms, so model names alone do not determine complexity or explanatory value. We therefore compare predictive performance alongside the interpretation each fitted pipeline supports.

The main question was how ranking changes from application-table information (A), to basic history (B), to behavioural enhancement (C) across three complete pipelines. Fixed original configurations and common folds evaluate each pipeline's information expansion. The secondary question concerns further gains from bounded LightGBM optimisation and fusion. Keeping these analyses separate distinguishes information gains from configuration gains within this retrospective development study.

2 Data, EDA and Feature Construction

2.1 Applicants, outcomes and missingness

The labelled data contain 307,511 applications, including 24,825 positive labels (8.07%). Each row represents an applicant identified by SK_ID_CURR; TARGET=1 denotes payment difficulty under the dataset’s outcome definition. IDs were used for joining and alignment, never as predictors. The official test set has 48,744 unlabelled applications, so its AUC cannot be computed locally.

Class imbalance motivated average precision (AP) alongside ROC AUC. Some application fields had substantial missingness. For example, COMMONAREA_MEDI was missing for 214,865 applicants (69.87%). Descriptive exploratory data analysis (EDA) used the labelled cohort and informed pipeline-specific missing-value handling. Aggregation retained every applicant, including those without matched history; learned preprocessing used only training portions.

2.2 Three source-defined information sets

Table 1 lists shared-table column counts, including ID and label, before model-specific encoding.

Table 1: **Feature sets**

Group	Information source	Columns
A	Application-table only, with four shared derivatives	126
B	Basic history: A plus 27 historical aggregates	153
C	Enhanced history: B plus 39 behavioural features	192

A retained the original application fields, including external scores and credit-inquiry counts already present in that table. It therefore means *application-table information*, not the complete absence of credit-history-related information. Shared deterministic derivatives were age and employment duration in years, income-to-credit ratio and annuity-to-income ratio. The special employment code 365243 was handled as missing/special status, not as an ordinary duration or a direct unemployment indicator.

B aggregates bureau, previous-application, instalment, point-of-sale and credit-card histories. C adds 39 repayment, recent-history, structural and monthly-bureau features. For example, B records mean instalment-payment amount; C adds the fraction of valid due loan-period events not fully paid on time, summing split payments. This fraction excludes events with missing payment information or conflicting schedules, while retaining the applicant. A recent version restricts valid due events to the preceding 365 days. No joined-history field, dependent ratio or history-derived missingness flag entered A. Full feature definitions remain in the notebook.

Before aggregation, the data audit excluded 17 bureau records updated after application, without changing raw CSVs or removing applicants. This specific check does not resolve every temporal ambiguity. Record counts need not count distinct loans. The legacy INSTALL_AVG_DAYS_LATE field means actual-payment-day offset, not delay from the scheduled date.

3 Modelling Pipelines and Evaluation

3.1 Fixed comparison and training boundaries

An ID-based, stratified split with seed 5054 assigned 292,135 applicants to development and 15,376 to a 5% holdout. All nine pipeline/information combinations used the same labels and three development fold assignments for cross-validation (CV). In each outer fold, the remaining two folds supplied training data. Imputation statistics, category vocabularies, scaling and learned basis transformations were fitted within the appropriate training portion, then applied to that fold’s validation data.

A was added after B/C results were known, requiring nine outer fits plus three internal early-stopping fits. The 18 historical B/C folds were authenticated through input, source, configuration, split and prediction checks. Saved A models were replay-checked; B/C outer models were not persisted.

Reproduction checks did not replace experimental results or reopen selection. The previously viewed holdout provides supplementary evidence only.

Figure 1 separates deterministic business aggregation and derivatives, which may precede splitting, from transformations learned only on training portions. Every information set uses the common ID split; outer validation receives fitted transformations without refitting. The holdout is outside this CV procedure.

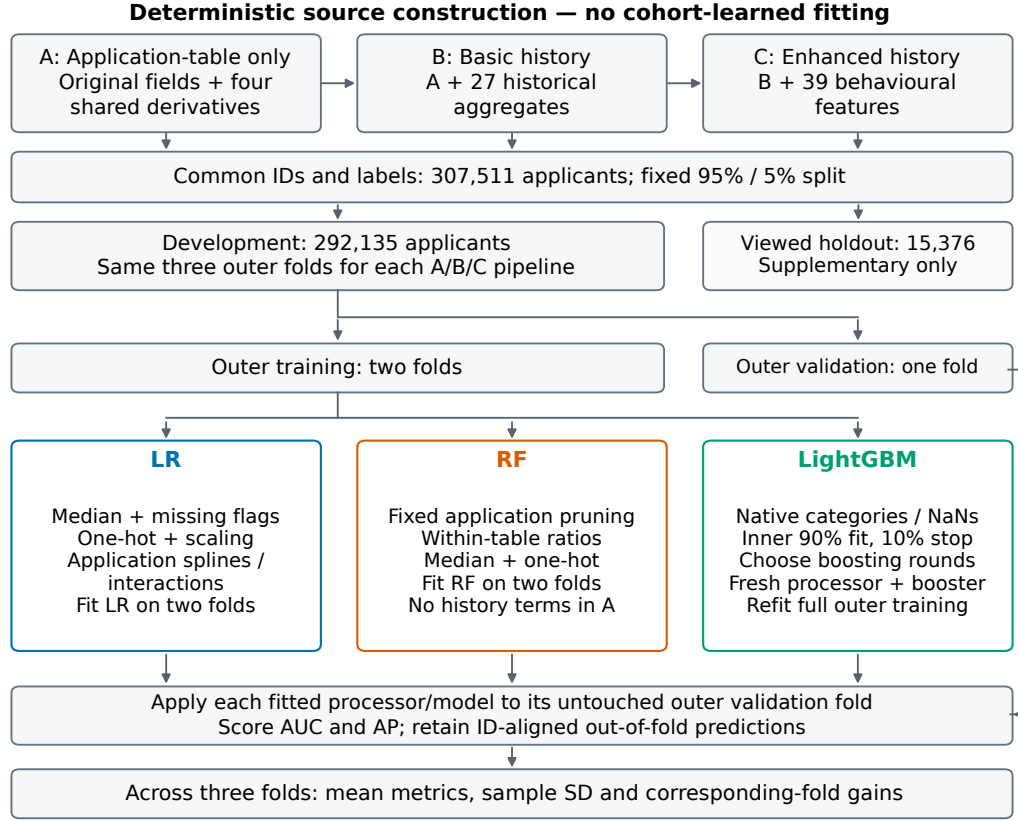


Figure 1: **Study design**

Logistic regression. LR used training-only median imputation, numeric missingness indicators, six predefined log transformations, full one-hot encoding and nonbinary scaling, with application-based splines and interactions. Fixed Elastic-Net settings were $C = 0.01$, L1 ratio 0.25 and balanced class weights. The numerical solver used equivalent coefficient splitting with bound-constrained optimisation and first-order optimality checks. Coefficients describe transformed terms, not necessarily an original variable's overall effect.

Random forest. RF used application-field pruning, within-application ratios, median imputation and one-hot encoding. History-dependent ratios and flags were excluded for A. Settings were 300 trees, depth 15, minimum leaf size 50, feature fraction 0.1, class weights 1:2, Gini criterion and seed 42. More columns increase the absolute number sampled per split, so information-set contrasts include induced representation changes.

LightGBM. The original configuration used native categorical and missing-value handling, depth 2, learning rate 0.1, at most 31 leaves, minimum leaf size 20 and balanced sample weights. A stratified internal 10% of each outer-training set, using seed 42, selected rounds by unweighted AUC with a 1,000-round cap and patience 50. The categorical processor and booster were then fitted afresh on the complete outer-training set for the chosen number of rounds. Outer validation labels did

not select stopping rounds. The later refined LightGBM configuration was excluded from A/B/C. Information-value (IV) screening was disabled in the formal unified experiments.

3.2 Metrics, pairing and secondary selection

ROC AUC is the area under the receiver operating characteristic curve and measures positive–negative ranking. AP summarises precision across recall levels under the observed class distribution. Both used the natural validation distribution without class weighting. The primary summary was the unweighted mean of three fold AUCs with sample SD (denominator 3 – 1); AP followed the same convention. Out-of-fold (OOF) predictions score observations excluded from the corresponding model’s training. We report the mean of fold AUCs rather than AUC calculated from pooled OOF predictions. Paired contrasts subtract corresponding folds before averaging. Because training sets overlap, fold SD describes variation across these folds and is not a confidence interval. We did not perform significance tests.

Secondary development selection on C covered 12 LightGBM configurations (eight structural/weight choices, four refinements), two extra feature groups and their combination over a frozen configuration range, and three prespecified fusions. The groups contain four recent–overall level differences and six Active/Closed-status delinquency summaries; overlapping windows make the former differences, not time slopes. Configuration selection retained candidates within 0.0003 of each pool’s maximum mean AUC, then applied a frozen complexity order. The order prioritised fewer added feature groups, shallower trees and fewer leaves; remaining tie-breakers are specified in the notebook. After combined-pool selection, an extra-feature winner required mean AUC gain > 0.0003 with no negative fold relative to the parameter-only choice. Fusion first applied this gain rule relative to selected LightGBM. Among eligible fusions within 0.0003 of the maximum, fewer components were preferred, then raw over rank fusion. These are development-selection scores, not unbiased estimates of the selected procedure.

4 Results

4.1 Gains from historical information

Historical information improved ranking in all three fixed pipelines (Figure 2). Basic history increased mean AUC by 0.009667 for LR, 0.011061 for RF and 0.014890 for LightGBM; the 39-feature stage added 0.011339, 0.003187 and 0.008138. Both stages increased AUC and AP on every recorded fold. Total C–A gains were 0.021006, 0.014248 and 0.023029. The first expansion benefited LightGBM most, whereas the second added most to LR. These contrasts support the incremental predictive value of joined history within each pipeline; they do not isolate any individual feature’s causal contribution. Table 2 reports all nine AUC/AP summaries as three-fold means $\pm$ sample SD, rounded to four decimals. A/B/C follow Table 1; reported differences were calculated before rounding. Full-precision fold scores remain in the notebook.

Table 2: Development performance

Pipeline	Metric	A: Application-only	B: Basic history	C: Enhanced history
LR	AUC	0.7529 $\pm$ 0.0025	0.7626 $\pm$ 0.0025	0.7739 $\pm$ 0.0027
	AP	0.2355 $\pm$ 0.0057	0.2472 $\pm$ 0.0045	0.2599 $\pm$ 0.0037
RF	AUC	0.7518 $\pm$ 0.0016	0.7628 $\pm$ 0.0010	0.7660 $\pm$ 0.0012
	AP	0.2340 $\pm$ 0.0050	0.2476 $\pm$ 0.0054	0.2534 $\pm$ 0.0044
LightGBM	AUC	0.7572 $\pm$ 0.0029	0.7721 $\pm$ 0.0028	0.7802 $\pm$ 0.0025
	AP	0.2420 $\pm$ 0.0051	0.2618 $\pm$ 0.0036	0.2722 $\pm$ 0.0021

Figure 2a uses circular markers for three-fold AUC means with SD on a zoomed axis. Panels b/c retain all 18 paired gains. Colours identify models throughout; only in b/c do circles, squares and triangles identify folds 0, 1 and 2. Mean segments and zero-inclusive paired axes expose fold consistency.

LightGBM had the highest mean AUC in every group. LR ranked above RF in A and C but below it in B by only 0.000206. This ordering is descriptive. Different representations and prior search effort prevent attributing performance differences to algorithmic complexity alone.

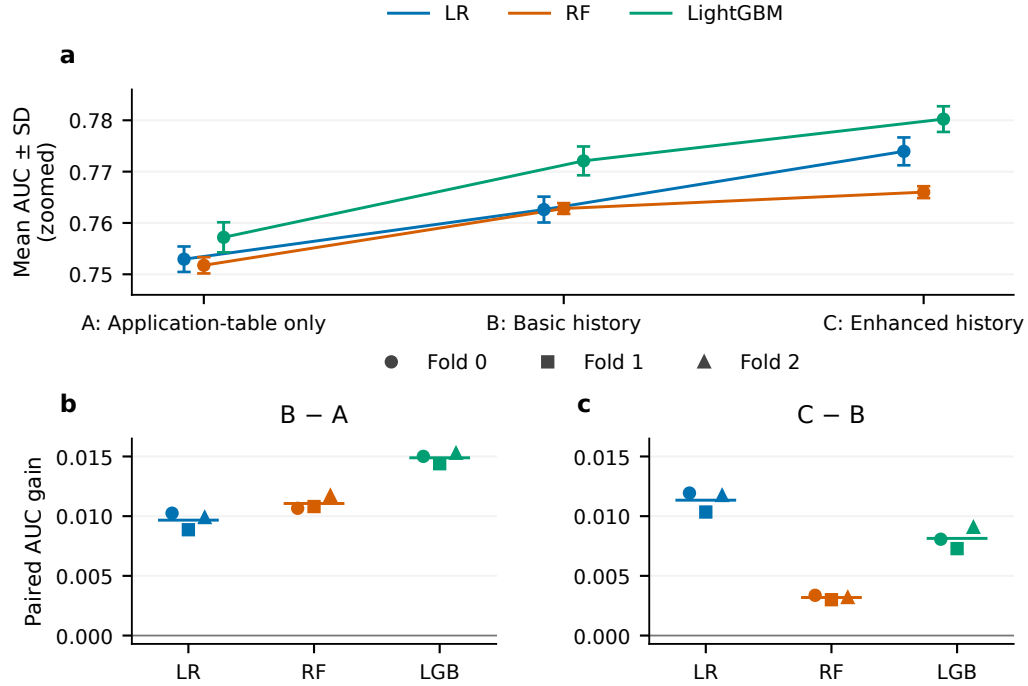


Figure 2: **Historical information gains**

4.2 Interpreting the fitted models

Figure 3 shows the five largest absolute coefficients or importances within each original C model fitted on full development data. The panels use separate scales: signed LR coefficients on transformed terms, RF impurity importance and LightGBM split gain in thousands. The tree models assign high importance to external scores already present in A; RF also uses an active net-debt-to-credit ratio. This is compatible with the A/B/C gains: importance describes a fitted model's use of correlated inputs, whereas the paired comparison evaluates an information-set expansion.

LR coefficients describe transformed terms, with mixed scaling, full dummy coding and nonlinear terms limiting interpretation at the original-variable level. A negative coefficient indicates a conditional association for that term; it does not establish that changing the original variable would reduce risk. RF impurity importance and LightGBM gain measure contributions to tree splitting without specifying effect direction, and their magnitudes cannot be compared across models. None of these diagnostics establishes causality, and cross-fold importance stability was not evaluated.

4.3 Bounded optimisation

With enhanced information held fixed, the final selected base configuration increased mean AUC from 0.780230 to 0.781855 (difference 0.001625). Refined LightGBM retained depth 2, but used learning rate 0.05, minimum leaf size 100, L2 regularisation 5 and unweighted training, with a 2,000-round cap and patience 100. AP increased from 0.272194 to 0.275559. This contrast changes configuration, not the information set. Figure 4 separates this contrast from the subsequent fusion comparisons.

The parameter-only rule initially selected the feature-trial reference (AUC 0.781613; gain 0.001383). Final selection reconsidered the combined configuration/feature pool. Its maximum, the combined extra-feature candidate, raised the tolerance floor to 0.781615, excluding that reference. Refined LightGBM then won the simplicity rule without adopting extra features. Thus its selection depends on the candidate pool; the stages are not independent. The notebook reproduces both frozen choices.

Extra-feature trials used the initial reference: depth 2, learning rate 0.1, minimum leaf size 20, L2 regularisation 5 and unweighted training. Recent-overall differences changed mean AUC by

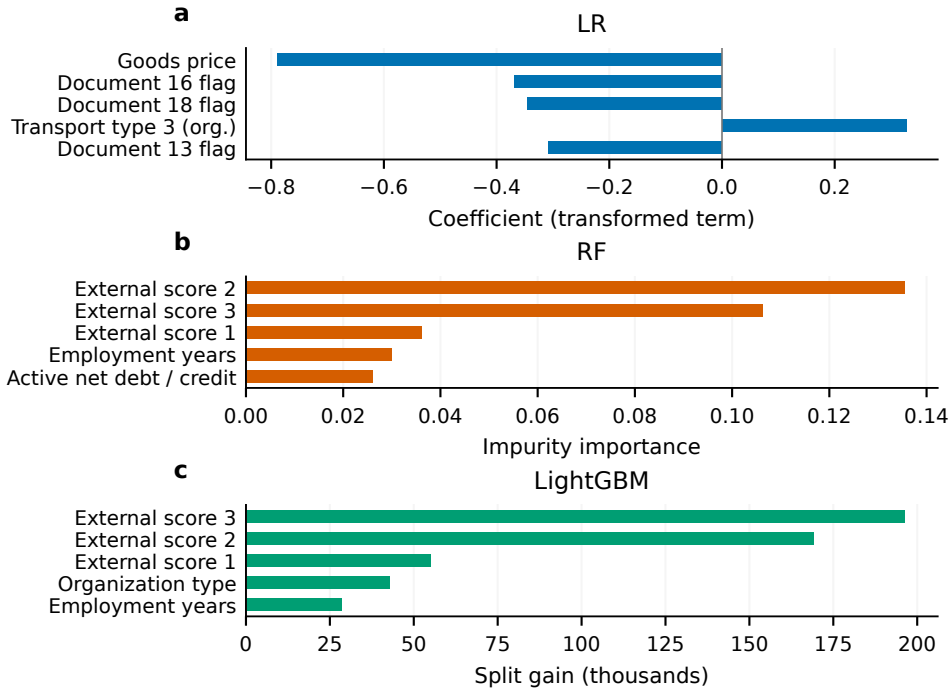


Figure 3: **Model diagnostics**

−0.000039 and status-specific summaries by −0.000109. Their combination gained 0.000303 but declined on one fold, failing the stable-gain rule. None was adopted. Sixteen optimisation development folds were flagged as possibly truncated; final internal runs reached the 2,000-round cap. Neither a performance ceiling nor full convergence is established. Complete candidates remain in the notebook.

4.4 Fusion trade-offs

Selected LGB–LR rank fusion (80%/20%) achieved AUC 0.782349: gains of 0.000494 over refined LightGBM and 0.002119 over original C LightGBM. Raw fusion at the same weights reduced AUC on every fold despite slightly higher mean AP (Figure 4). Selected rank fusion increased mean AP only slightly and had two negative AP folds. Adoption targeted AUC, not improvement of every metric.

LGB–LR–RF rank fusion (70%/20%/10%) had higher mean AUC, 0.782638. Its 0.000289 advantage lay within the frozen 0.0003 tolerance, so the simpler two-model fusion was selected. The frozen rule thus selected the simpler fusion despite its lower observed mean AUC.

Figure 4 shows paired AUC (a) and AP (b) changes on common development folds. The top row compares refined with original C LightGBM; the lower rows compare fusion with refined LightGBM. Weights follow model-label order. All folds, mean segments and zero are shown, without intervals. Extra-feature trials use a different reference and remain separate.

4.5 Supplementary evaluation

Table 3 reports AUC/AP on the previously viewed 15,376-applicant holdout. All five pipelines use C; the first three retain their original configurations. CDF fusion increased AUC over refined LightGBM but reduced AP. This single, previously examined split does not provide independent confirmation, and it prompted no further selection.

Fusion cross-validation used percentile ranks computed inside each validation batch. Holdout and official-test scoring instead mapped model scores through development-OOF empirical cumulative

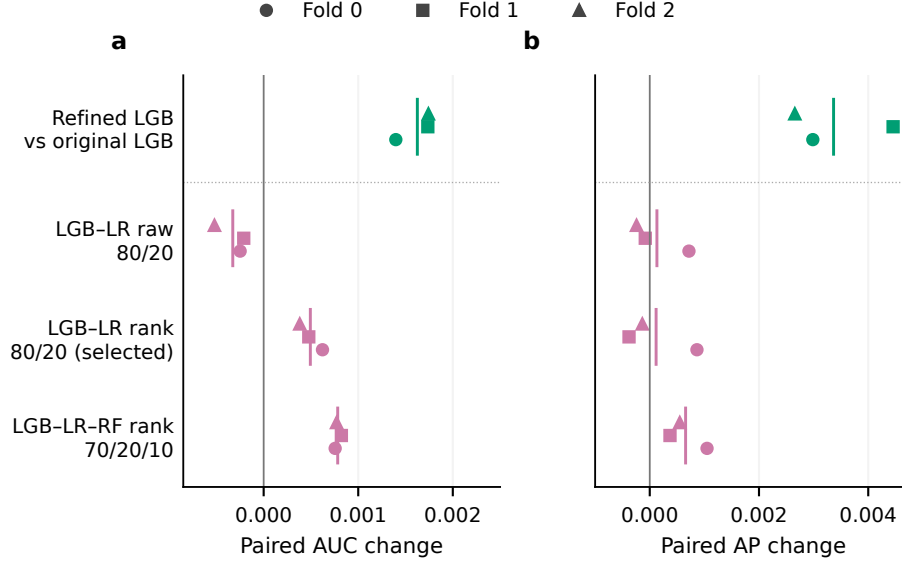


Figure 4: **Refinement and fusion**

distribution functions (CDFs), giving LGB-LR CDF fusion with the same 80%/20% weights. These are different scoring rules; the CV batch-ranking gains do not validate equivalent deployment-CDF performance. Fusion outputs are ranking scores, not calibrated probabilities of payment difficulty.

An author-provided Kaggle screenshot for the submission labelled “LGB-LR rank 80/20” reports a score of **0.77948** and a private score of **0.77738**, with status “Complete (after deadline)”. These supplementary scores prompted no model reselection, and no official leaderboard rank is reported. The supplied screenshot links the scores to the submission description; the uploaded file was not downloaded for byte-level matching.

Table 3: **Viewed holdout**

Pipeline	AUC	AP
LR	0.773467	0.249358
RF	0.769716	0.250608
Original LightGBM	0.778127	0.266457
Refined LightGBM	0.779527	0.269148
LGB-LR CDF fusion (80/20)	0.780285	0.266881

5 Discussion

The contrast between useful historical expansions and rejected later additions shows that feature value depends on the starting representation and evaluation criterion. More candidate features did not ensure consistent gains, even when their average gain was positive. This favours judging additions by observed validation benefit and fold behaviour within a specified pipeline. Diminishing returns remain one interpretation of the bounded search; the results establish neither feature redundancy nor a ceiling for feature engineering.

The differing gains may reflect representation as well as information. Explicit repayment summaries could make useful relations more accessible to LR; RF’s smaller second-stage gain could reflect existing splits or its fixed depth and feature sampling. No component ablation distinguishes these explanations, so neither redundancy nor algorithmic capacity is established as the cause.

Predictive performance and explanatory usefulness also require separate assessment. LR’s mathematical structure is inspectable, but its transformations restrict simple coefficient interpretations. Tree importances identify frequently useful inputs without explaining effect direction or individual decisions. LightGBM’s stronger observed ranking therefore does not establish superior explanatory or operational value. Calibration, fairness and lending-policy utility were not evaluated.

Selection dependence limits generalisation. A was retrospective, development data supported exploration and secondary selection, and the 5% holdout had already been viewed. No complete nested evaluation or fresh external test removes this dependence. Overlapping training folds provide descriptive variability, while the batch-rank/CDF distinction limits transfer of fusion gains. Historical-information gains exceeded later optimisation gains here, but unequal search budgets and pipeline choices prevent generalising this ordering to feature engineering versus tuning.

6 Conclusion

Under fixed original pipelines and shared development folds, both historical-information stages improved payment-difficulty ranking beyond application-table information. The gains varied across pipelines, while bounded LightGBM optimisation and fusion produced smaller additional gains with AUC/AP trade-offs. The study supports evaluating information additions separately from configuration changes and retaining negative feature trials in the comparison. These conclusions apply to the recorded development experiment; independent evaluation is still needed to establish generalisation and deployment benefit.

Contributions and AI Use

WU Ziyi was responsible for the data preparation and EDA module; JIANG Zerui handled the LR baseline and team/report integration. MA Ruiyang handled RF, and CHEN Kepeng handled LightGBM. These module responsibilities include work completed with the AI assistance disclosed below.

AI and large language models (LLMs) assisted code inspection, repairs, experimental validation, A/B/C supplementation, bounded optimisation, figures, notebook integration and manuscript preparation. Final report review added no model fits, candidate configurations or model selection. The authors remain responsible for the accuracy, interpretation and submission of the work. The accompanying notebook records provenance and recomputes saved-prediction metrics with zero fits under default Run All. Raw reconstruction and frozen training require separate commands and original data; VERIFICATION.md distinguishes current checks from historical runs. The recorded environment used Python 3.13.9, scikit-learn 1.7.2 [5] and LightGBM 4.6.0.

References

- [1] Home Credit. *Home Credit Default Risk*. Kaggle competition dataset. <https://www.kaggle.com/competitions/home-credit-default-risk>.
- [2] H. Zou and T. Hastie. Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society, Series B*, 2005. <https://doi.org/10.1111/j.1467-9868.2005.00503.x>.
- [3] L. Breiman. Random forests. *Machine Learning*, 45:5–32, 2001. <https://doi.org/10.1023/A:1010933404324>.
- [4] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye and T.-Y. Liu. LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems*, 30, 2017. <https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html>.
- [5] F. Pedregosa et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011. <https://www.jmlr.org/papers/v12/pedregosa11a.html>.